

How TECHNOLOGY Is Helping SPECIALLY-ABLED PEOPLE



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This year, the government of India dedicated the National Science Day to the theme “science and technology for specially abled persons.” The idea was to encourage innovation, awareness and reach of accessibility solutions to specially-abled people in the country.

The differently-abled population in India is estimated to be 100 million, half of which is under the age of 30. This indicates a huge need for assistive technology products. The government plans to work together with corporates, educators, NGOs and the differently-abled people so as to promote availability of solutions that can help differently-abled people to lead independent lives.

People with disabilities meet barriers of all types. However, with the help of assistive technology, they are now able to do things that would have never been possible before—from switching on a light to having a voice to express themselves. Technology has always lent a helping hand for people with visual impairment, speech impairment and motor disabilities. The mobile technology, in

particular, has provided the foundation for development of a lot of apps and gadgets that can help ease the difficulties people with disabilities face on a daily basis.

Devices for people with visual impairment

Dot. It is the world’s first Braille smart-watch. A practical, wearable solution, it is more affordable than regular e-Braille devices which may cost thousands. Dot helps the visually impaired to access messages, tweets and even books anywhere and at any time. It can connect via Bluetooth to any smartphone, then retrieve and translate the text (from an email or messaging app) into Braille for its owner.

Braille ebook reader. A Kindle-style e-reader, it lets visually impaired people read easily. Alphabets are represented as raised bumps. So the visually impaired can read by tracing lines of raised bumps with the help of their fingertips. The reader also helps to understand graphics and figures.

Tactile wand electronic stick. It lets visually impaired people to determine the objects that come in their way. As soon as the user approaches an object, the stick starts vibrating. The closer the user comes to the object, the greater the vibration.

FingerReader. This wearable tool assists in reading as well as translating text. The user can wear the device on a finger, then point it to a body of the text, one line at a time. The small camera on the FingerReader scans the text and gives real-time audio feedback of the words it detects. It also notifies the reader via vibrations when the user is

FingerReader for the visually impaired (Image courtesy: news.mit.edu)

